THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 1: **Exploring the Investigative World of Science** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A table of observations helps us to: 1

(a) skip measurements **(b)** change results
(c) organise data and see patterns **(d)** avoid thinking

2. While heating a liquid in a test tube, the mouth of the tube should point: 1

(a) towards a friend **(b)** away from you and others
(c) towards the ceiling lamp **(d)** towards your face

3. Which of the following is a testable scientific question? 1

(a) Does the amount of water in the dough change how well a puri puffs?
(b) Is cooking a boring activity?
(c) Who will eat the most puris?
(d) Which puri looks nicest?

4. A graph is drawn from data mainly to: 1

(a) make the data longer
(b) replace measurements
(c) hide the numbers
(d) show the relation between two quantities clearly

5. Wearing safety goggles in a laboratory protects the: 1

(a) ears **(b)** hair only **(c)** eyes **(d)** feet

6. Recording whether a puri puffed (yes or no) is an example of: 1

(a) qualitative observation **(b)** quantitative measurement
(c) a hypothesis **(d)** a conclusion

7. Which of the following is NOT good scientific practice? 1

(a) repeating a test
(b) sharing results with others
(c) recording every observation honestly
(d) changing the data so that it matches the hypothesis

8. An experiment in which the effect of only one factor is studied while others are kept constant is a: 1

(a) fair test **(b)** random test **(c)** survey **(d)** guess

9. Recording that a puri puffed in 8 seconds is an example of: 1

(a) quantitative observation **(b)** a question
(c) an opinion **(d)** qualitative observation

10. A beaker of hot water is touched to check its temperature. A better, safer and more reliable method is to use: 1

(a) the tongue **(b)** a spoon **(c)** the thumb **(d)** a thermometer

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Name two instruments used in the puri-puffing investigation and state what each measures. 2

12. What is meant by a conclusion? On what should it be based? 2

13. Why must only one factor be changed at a time in a fair test? 2

14. Why should results never be changed to match the expected answer? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain the difference between an observation and an inference with an example. 3

16. A student writes: 'Hot oil makes puris puff.' Rewrite it as a testable hypothesis and name the variables. 3

17. An experiment produces a result that does not support the hypothesis. List what the scientist should do. 3

OR

Why is it not enough to do an experiment only once? What should be done to improve reliability? 3

18. Why do scientists share their findings with others? Give two benefits. 3

OR

Explain why 'the puri is crisp' is an observation while 'the oil is hot enough' is an inference. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Differentiate between qualitative and quantitative data, giving two examples of each from cooking. Explain which is more useful for comparison. 5

OR

Explain why a conclusion must be supported by data. What may happen if people accept claims without evidence? Give an example. 5

20. Heights of a plant after one week under different amounts of water are 5, 8, 11, 11 and 10 cm. Find the mean and mention how an unusual value should be treated. 5

OR

Explain the terms: hypothesis, variable, observation, inference and conclusion with one example each from an investigation of oil temperature and puri puffing. 5

END OF QUESTION PAPER — Total: 40 Marks